Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3453**

Title: **GLOBAL BUCKLING OF FLAT COMPOSITE STIFFENED PANELS**

Revision: **0.2**

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1 INTRODUCTION

This chapter discuss the available methodologies for calculating the buckling onset of flat composite stiffened panels. Airframes of modern aircraft often contain this type of structure to configure upper and lower panels of lift boxes (wing and tail planes). Global (or general) buckling, in opposition to local buckling, refers to buckling modes in which the stiffening members of the panel are not fully effective as supports for the skin, and buckling waves are capable to push them away from their original location.

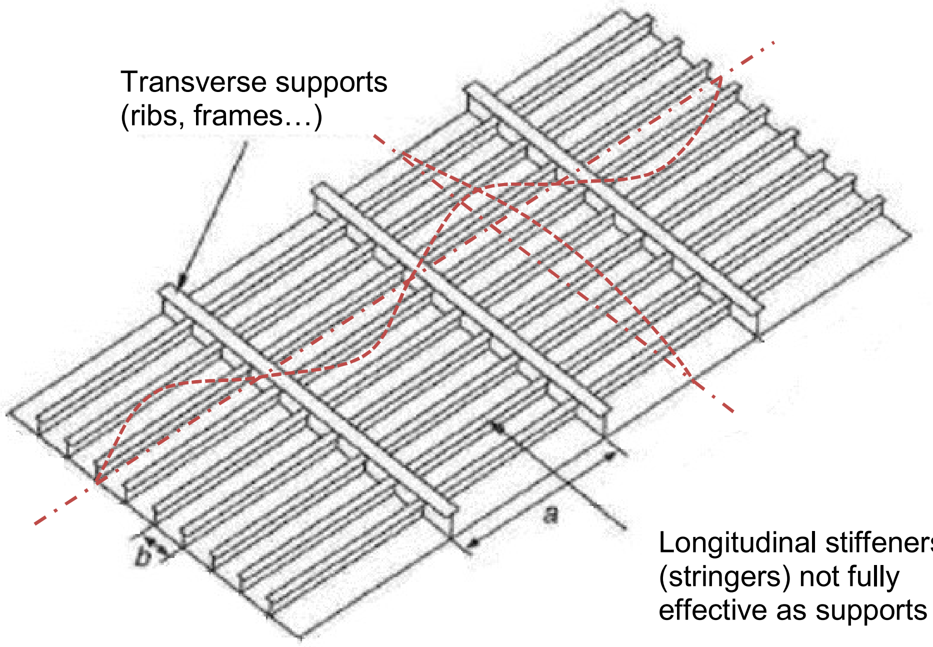

Figure 1. Flat stiffened panel and sketch of a global buckling mode

2 HIGH FIDELITY METHODS

2.1 FINITE ELEMENTS MODELS

The standard FEM solvers used within Airbus (including MSC Nastran) include the buckling analysis as a variant of the linear static solution (elastic materials, small displacements) frequently used in stress justification. This analysis option of the FEM solvers are capable to calculate with precision the buckling onset (local and/or global) provided the finite element model is built consistently with the kind of structure analysed and the objective searched. Some typical rules and recommendations to create a FEM of a composite stiffened panel intending to get its buckling loads are (not exhaustive, see Ref. 9 and other FEM modelling guidelines for further advice):

- ✓ Macroscopic models, with the composite laminates idealized as 2D shell elements, are sufficient
- ✓ The aspect ratio of the 2D elements, i.e. the ratio between its maximum and minimum dimensions, should be not be far from 1.0 (the square shape)
- ✓ If both local and global modes should be captured, the mesh will be prepared in a manner that panel bays between stiffeners will contain at least 10 elements, and flanges with and edge supported and the other free (eg. the web of a tee stringer) will contain at least 5 elements in the section
- ✓ Variants of the modelling strategy may consider the stiffeners as beams (1D elements), if local buckling modes of the stiffener section are not object of the search (see figure below)

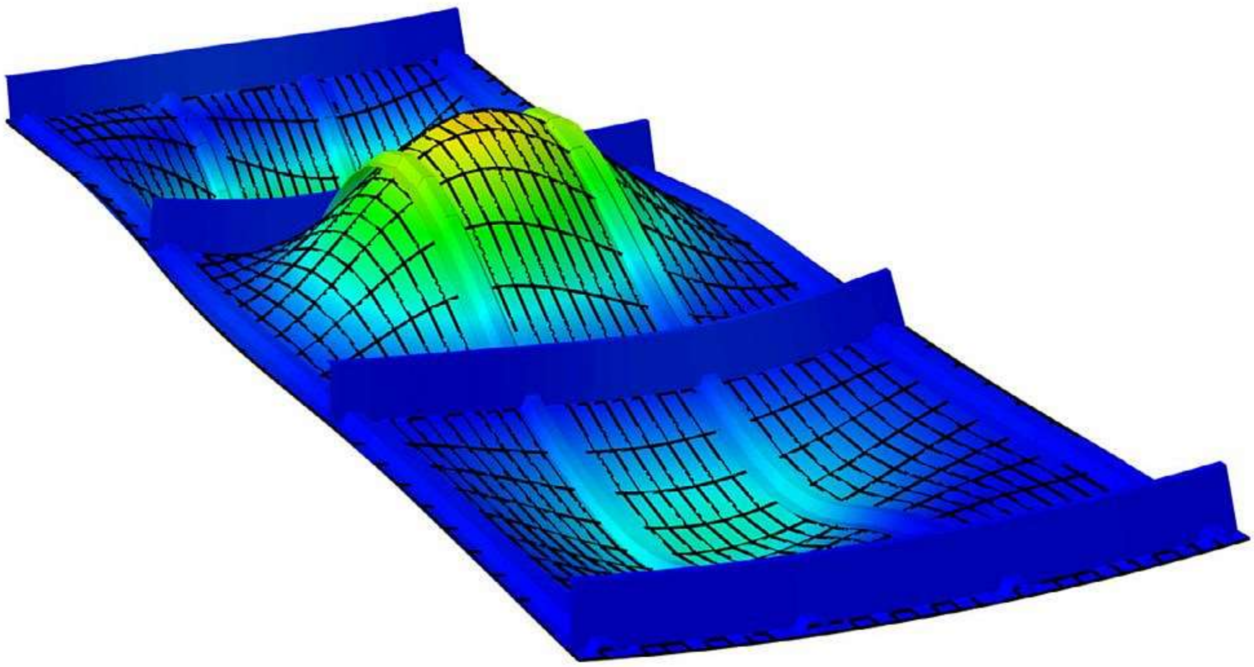

Figure 2. Global buckling of a stiffened panel obtained by FEM analysis

This methodology can capture both local and global buckling modes (provided the modelling strategy considered that objective). Differentiation is typically done by visual inspection of the buckling shapes in a FEM post-processing software.

2.2 ENERGY RAYLEIGH-RITZ METHODS

Another high-fidelity approach for predicting the buckling onset of structures is the Rayleigh-Ritz energy method. It has been used historically within Airbus for some structures. Each structural configuration needs a dedicated development, but avoids the dependencies on Finite Element solvers. The general idea of the method is to propose a series of form functions for the deformed (buckled) structure, compliant with the boundary conditions of the problem and find the solution by minimizing the total energy (potential plus deformation energies). The buckling problem is transformed into an eigenvalues calculation of some dense matrices after some mathematical work (see references Ref. 3 to Ref. 7 for further details).

As in FEM based analysis, this methodology can capture local and global buckling modes. Dedicated analysis of the buckling waves (eigenventors) shall be needed to differentiate. The criterion used for doing this classification automatically is setting, as convention, a maximum out-plane displacement of the stiffener as the limit to consider that the type of buckling was local. A typical convention for this threshold is 0.1 times the maximum out-plane displacement of the wave at any point of the panel.

This method is available, via dedicated software, within Airbus DS, for composite rectangular flat stiffened panels (see Table 1).

2.2.1 Stiffeners in the buckling analysis

Within these methods, the stiffeners are treated as 1D elements (straight beams), with cross sections idealized per the cross-section inertias: the axial stiffness (*EA*), the position of the "centre of gravity", the bending stiffness relative to an axis parallel to the skin (*EI*), and the torsion stiffness (*GJ*). A consequence of this simplification is that failures associated to local buckling of the stringer segments will not the captured by this method (further analysis, eg. the column buckling analysis of the super-stringer element would cover this limitation).

The reference axis system of the problem is positioned in the skin mid-plane for practical reasons, and the stiffeners (e.g. stringers) installed at one side of the skin (inner side in a lift box). Then, the *EI* of the stringer (normally available referred to its own "centre of gravity") must be introduced into the problem equations corrected by this offset (distance from stringer CG to the skin mid-plane) using the familiar Steiner's theorem.

A further correction used in Airbus DS methods for one-sided stringers is that proposed by the NACA report TN 2873 (Ref. 8). Transformed to a convenient notation for the problem of the composite panel stiffened with longitudinal stringers, the effective flexural stiffness of the one-side stringers to consider, shall be:

$$EI_{ef} = EI + \frac{EA \cdot \left(e_o + t_p - \frac{t}{2}\right)^2}{1 + Z_n \frac{EA}{E_x b t}}$$

Where:

EA	Axial stiffness of the stringer cross section
EI	Bending stiffness of the stringer cross-section, relative to its "centre of gravity"
e_0	Distance from the stringer "centre of gravity" to the skin surface
t	Thickness of skin

t_p Thickness of skin under stiffener (pad thickness)

b Distance between stringers

E_x Elastic modulus of skin in longitudinal direction

Z_n Factor per graph below

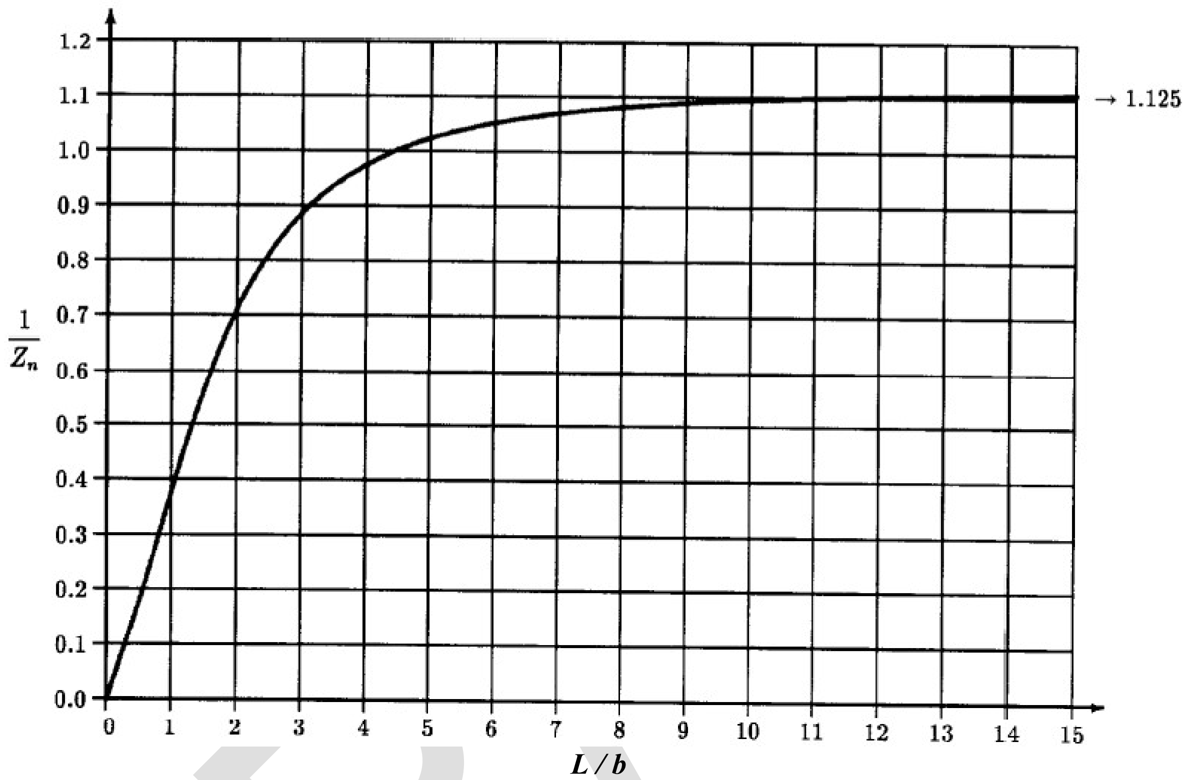

Figure 3. Effective flexural stiffness of stringers per NACA TN 2873 (Figure 3, N = ∞)

With this correction, the effective bending stiffness of stringers, once translated to the skin mid-plane can be typically of about 0.7 – 0.9 times the value obtained by literal application of the Steiner's theorem.

Note: The "Centre of Gravity" of the stringer cross section referred in this chapter is really the beam axis in axial and bending loads, calculated by weighting the segments areas with the laminates moduli, i.e. the result of $\sum(EA_i \cdot x_i) / \sum(EA_i)$, with EA_i and x_i the axial inertia and the geometrical centre, respectively, of each of the segments of the cross section.

3 APPROXIMATED METHODS

3.1 COLUMN BUCKLING OF A SUPER-STRINGER IN COMPRESSION

The column buckling stability of the beam composed by the stringer plus the corresponding effective skin, of length that between the consecutive transverse supports (ribs, frames…), under compression, shall be checked as per method defined in chapter §3440 "COLUMN BUCKLING OF COMPOSITE BEAMS".

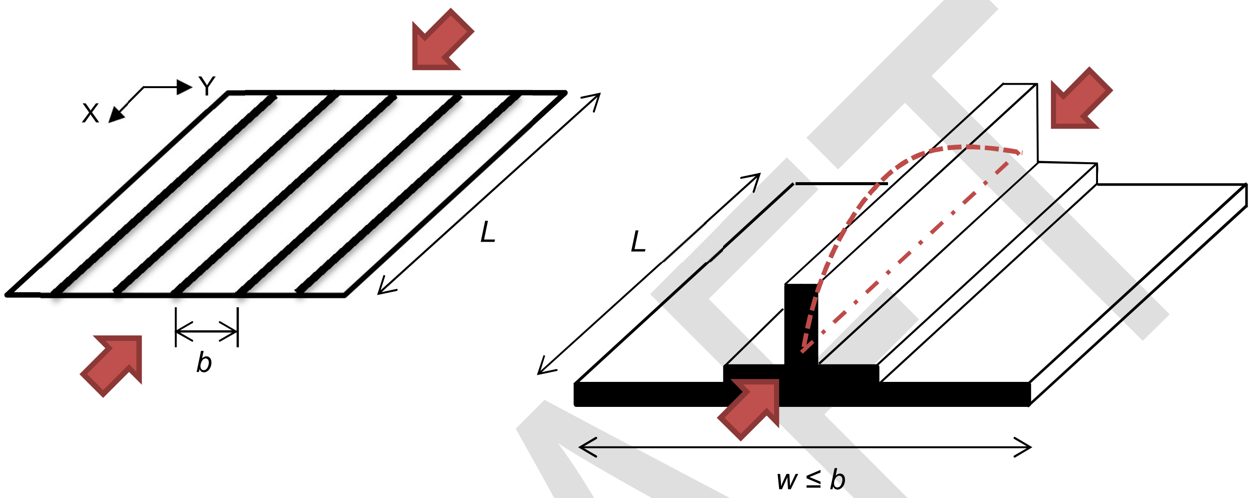

Figure 4. Super-stringer as column

The stringer crippling stress (and the inter-rivet buckling if applies) shall be used as cut-off value for the Johnson-Euler buckling calculation. Note that the calculation process of the super-stringer column is iterative, as the effective width is function of the longitudinal stress (i.e. the stress at the column buckling load which is searched). Other chapters relevant for this analysis are, then:

- §3430 "LOCAL BUCKLING AND CRIPPLING OF BEAMS/STRINGERS"
- §3480 "INTER-RIVET BUCKLING"
- §3485 "EFFECTIVE SKIN WIDTH"

This kind of allowable load, and its associated strength analysis, is usually added to complete the predictions of the high-fidelity methods discussed previously, as it takes into account, up to some extent (eg. risk of stringer disbonds are not covered), the post-buckling capability of the structure in longitudinal compression.

3.2 THE "EQUIVALENT PLATE" METHOD

The use of smeared properties for the assembly consisting of panel plus its stiffeners, beyond the column buckling analysis discussed previously, for calculating is general/global buckling loads, is an idea discussed (at least) by Timoshenko (see Ref. 1 chapter §9.11) and made explicit for composite stiffened panels by Kassapoglou (Ref. 2, chapter §9.1). The global buckling loads of the stiffened panel is approximated by the solution for buckling of an equivalent laminated plate made of a fictitious laminate with some stiffness matrices representative of the whole assembly (skin and stiffeners).

If $[D]_{sk}$ is the (generalized) bending stiffness matrix of the skin of that stiffened panel, and *EI* (*), *GJ* the bending and torsional inertias of the (longitudinal) stiffeners, separated by a distance *b* , the stiffness matrix *[D]* of that equivalent plate would be that same of the skin except for the following terms:

$$D_{11} = D_{11\text{-}sk} + EI / b$$

$$D_{66} = D_{66\text{-}sk} + GJ / (2 \cdot b)$$

(*) The stringer *EI* must be referred to an axis system at the skin mid-plane (see §2.2)

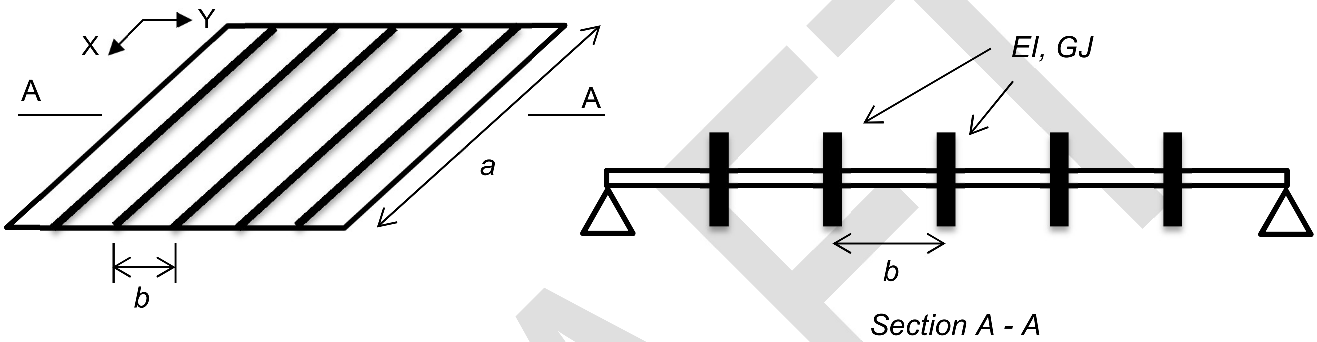

Figure 5. Flat rectangular panel with longitudinal equally-spaced stringers

With the stiffness matrix *[D]* of the equivalent plate, it is possible to apply the solutions for buckling in chapter 3450 “BUCKLING OF COMPOSITE PLATES”. The result would serve as approximation of the global buckling load of the stiffened panel.

3.2.1 Critical buckling load of the stiffened panel

The critical buckling load of the stiffened panel, no matter what is the type of buckling mode, could be then obtained as the minimum of these 2 values:

- ✓ Local buckling of the skin bay, from solutions for plate buckling using the skin laminate and a width equal to the distance between stiffeners (assuming they provide sufficient support)
- ✓ Global buckling of the stiffened panel, from solutions for plate buckling using the equivalent laminate described above (with smeared properties of skin plus stringers) and width not less than the length *a* or a minimum number of repetitions of the super-stringer element (typ. ≥6 skin bays)

The results of this approximation (envelope Local-Global) has shown a reasonable accuracy, if compared with the solution obtained from high-fidelity methods (Rayleigh-Ritz), in axial compression load and in “edge cases” of in-plane shear…

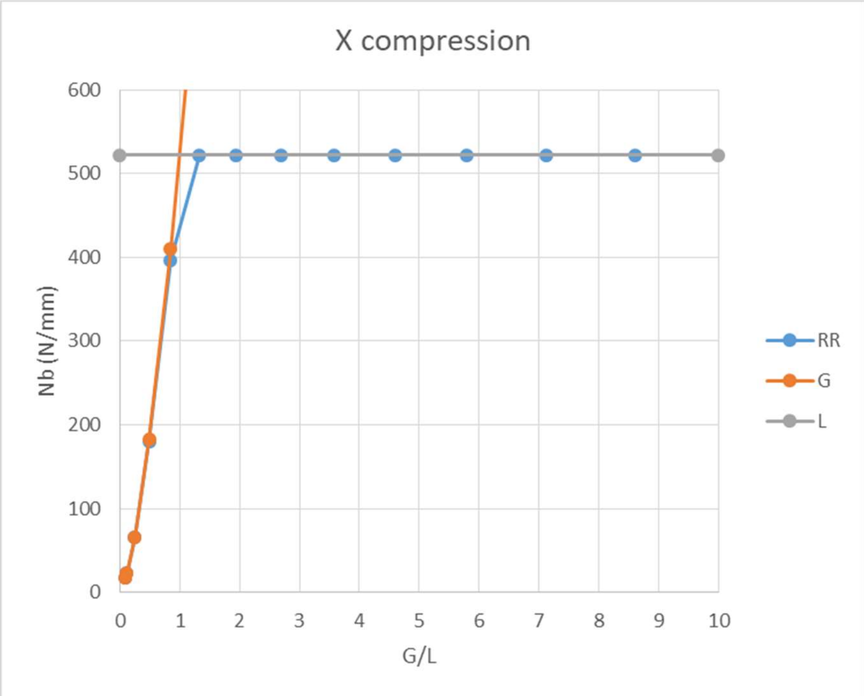

Figure 6. Envelope Local-Global vs high-fidelity methods in longitudinal X-compression

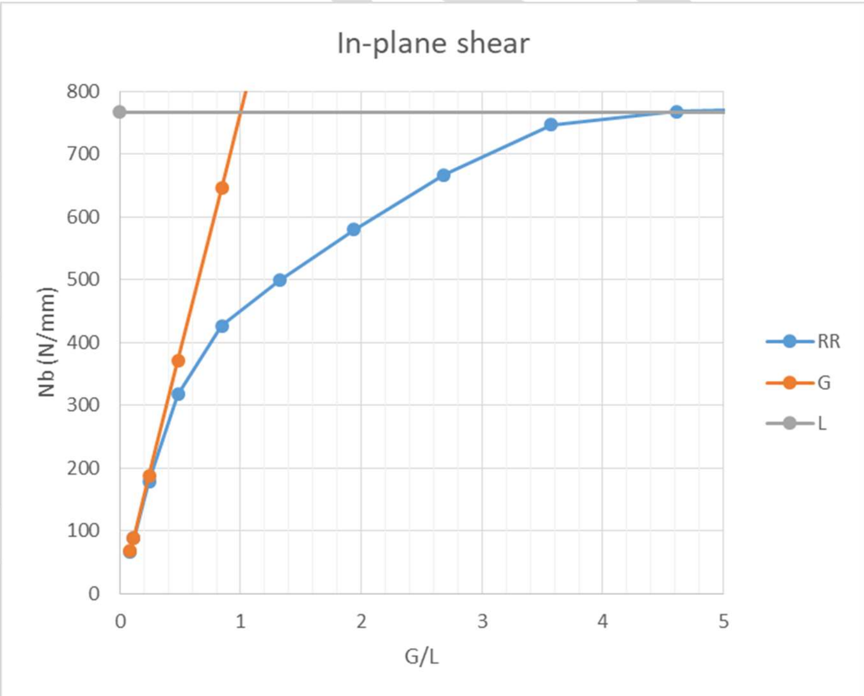

Figure 7. Envelope Local-Global vs high-fidelity methods in in-plane shear

… While when the stiffened panel is subjected to in-plane shear, some interaction between the types of buckling occurs, if the structural configuration is not clearly prone to skin buckling (high stringer *EI*) or to global buckling (low stringer *EI*). Note the parameter *G/L* in previous graphs represent the ratio between the global buckling load (calculated with the “equivalent-plate” method) and the local skin buckling load (assumed effective stringers).

A simple correction which respect the trends at the edge points, which capture the abovementioned coupling of buckling modes up to some degree (still imprecise/optimistic within that intermediate range) could take the following form:

$N_b = N_{b\text{-}sk} \cdot [1 - 1 / e^{G/L})$ where: $G/L = N_{b\text{-}gl} / N_{b\text{-}sk}$

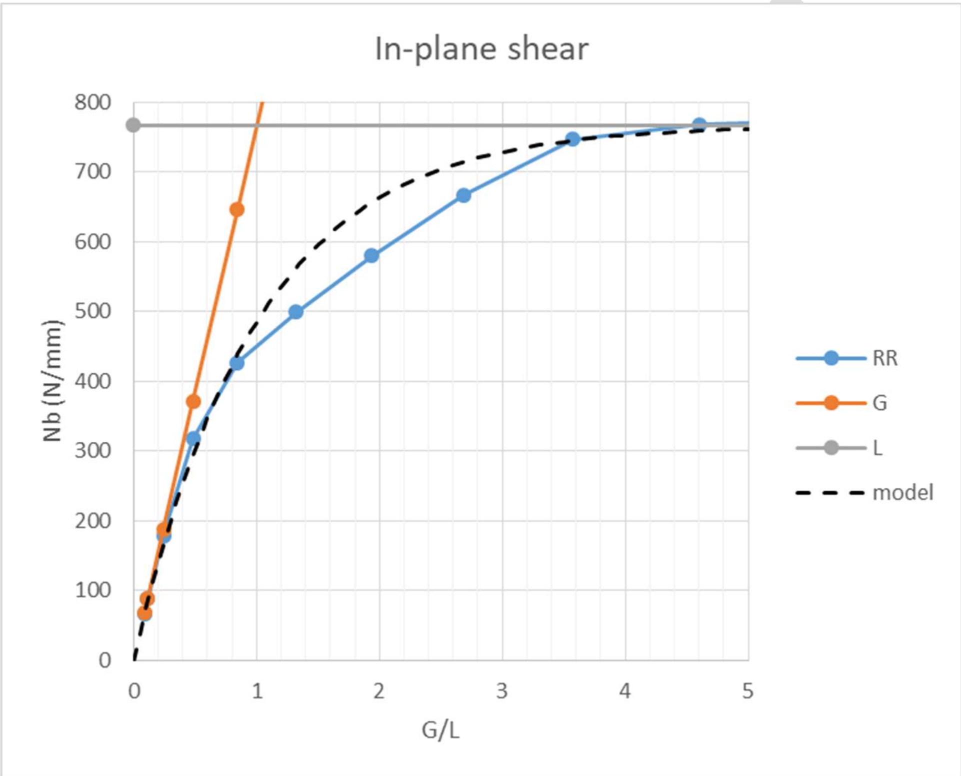

Figure 8. Correction for the approximated method for in-plane shear

Note that the local buckling load of the skin, in longitudinal X-compression, to be used in those expressions, must be expressed consistently with the global buckling loads (i.e., in terms of total axial force in panel divided by panel width). That is, to assure X-strain compatibility at stringers and skin, being *EA* the axial stiffness of the stringers, and $[A]_{sk}$ the in-plane stiffness matrix of the skin laminate, then the corrected local buckling of the panel, from the skin plate solution, shall be:

$N'_{bx\text{-}sk} / = N_{bx\text{-}sk} \cdot (A_{11\text{-}sk} + EA / b\ ) / A_{11\text{-}sk}$

<u>Don’t use this simplified approximated methodology for formal strength check substantiations.</u>

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
a	mm	Longitudinal distance between transverse supports of a panel
[A]	N/mm	In-plane stiffness matrix of a laminated composite
b	mm	Distance between consecutive stringers (width of skin bay)
[D]	Nmm	Bending stiffness matrix of a laminated composite
EI	Nmm^2	Bending stiffness of a stringer
G	-	Refers to General or Global buckling
GJ	Nmm^2	Torsional stiffness of a stringer (across an axis parallel to skin)
L	-	Longitudinal distance between transverse supports of a panel (as a), or, from context, refers to the Local buckling of the skin
N_b	N/mm	Buckling load (per unit width)
w	mm	Effective width of skin

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[Stephen P. Timoshenko]	2nd Ed.	THEORY OF ELASTIC STABILITY
Ref. 2	[Christos Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures
Ref. 3	[Airbus DS] NT-T-ADP-06002	Ed. A	Manual teórico del programa PANPA, versión 5.0
Ref. 4	[Airbus DS] TEA-T-NT-210048	Ed. B	Manual Teórico del Programa ARPA
Ref. 5	[Airbus DS] NT-T-ADP-00005	Ed. A	MANUAL TEORICO DEL PROGRAMA PANSA, Versión 4.0
Ref. 6	[J. N. Reddy]	2nd Ed.	MECHANICS of LAMINATED COMPOSITE PLATES and SHELLS. Theory and Analysis.
Ref. 7	[Airbus DS] XXXX	XXX	Stiffened Panel Analysis using Energy Method – SPAEM
Ref. 8	[P. Seide, NACA] TN 2873	-	THE EFFECT OF LONGITUDINAL STIFFENERS LOCATED ON ONE SIDE OF A PLATE ON THE COMPRESSIVE BUCKLING STRESS OF THE PLATE-STIFFENER COMBINATION
Ref. 9	[Airbus DS] NT-T-ADP-10003	Ed. B	Guía para la realización de Análisis de Estructuras mediante Modelos de Elementos Finitos de Detalle

SOFTWARE

Program	***Version***	***Description***
NASTRAN [MSC Software]	≥ 2013	Finite element analysis
PANPA [Airbus DS]	5.8	Buckling of flat unstiffened/stiffened rectangular panels
ARPA [Airbus DS]	5.1	Strength check of flat composite stiffened panels of lifting surfaces
SPAEM [Airbus DS]	X.X	Stiffened Panel Analysis using Energy Method

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.2 13/07/2022	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños DD/MM/2022	F. Glock DD/MM/2022

Table 3. *Approval signatures table for last revision of this chapter*